JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SHRI MURUGAN AUTO FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

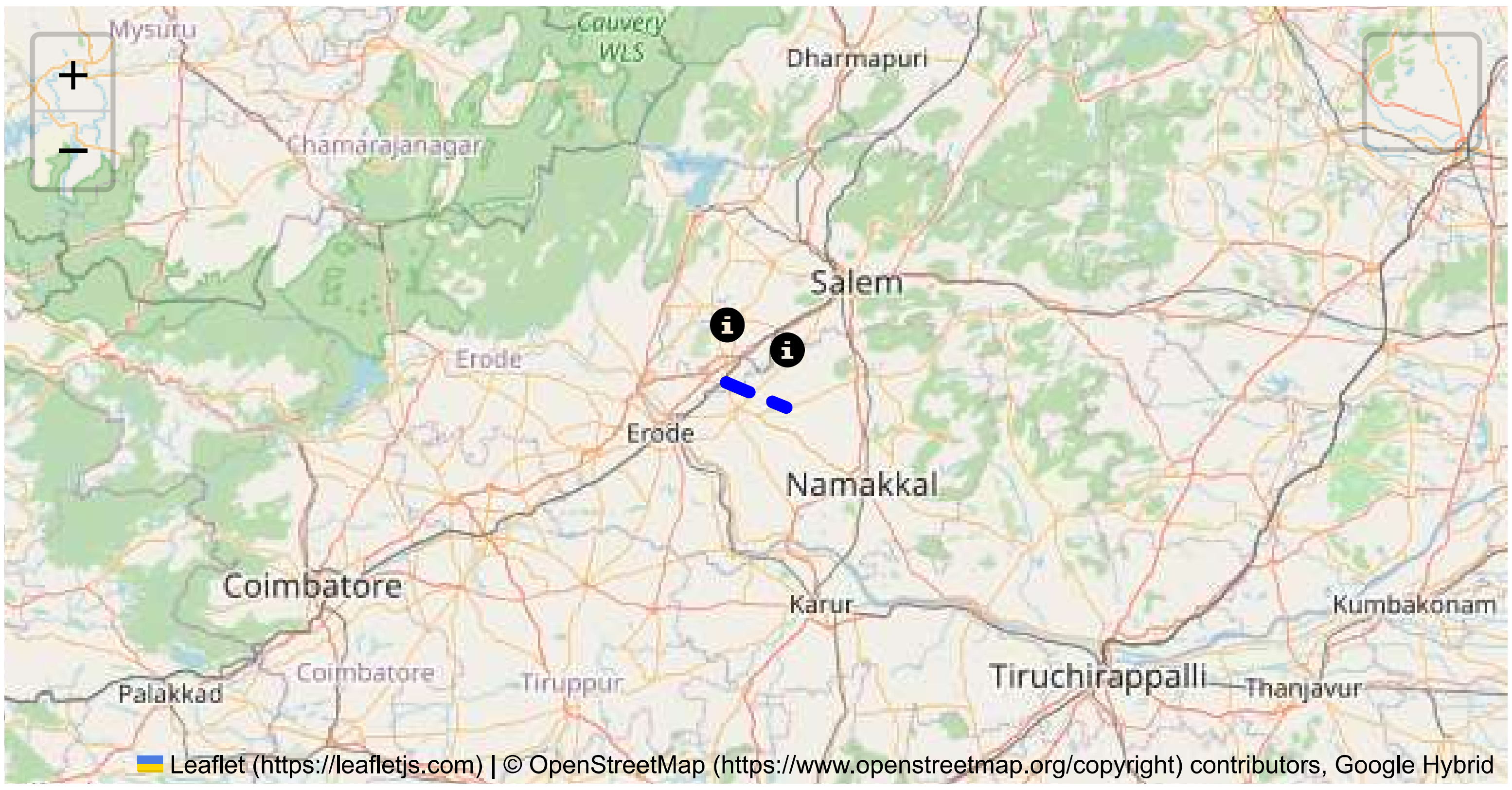
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

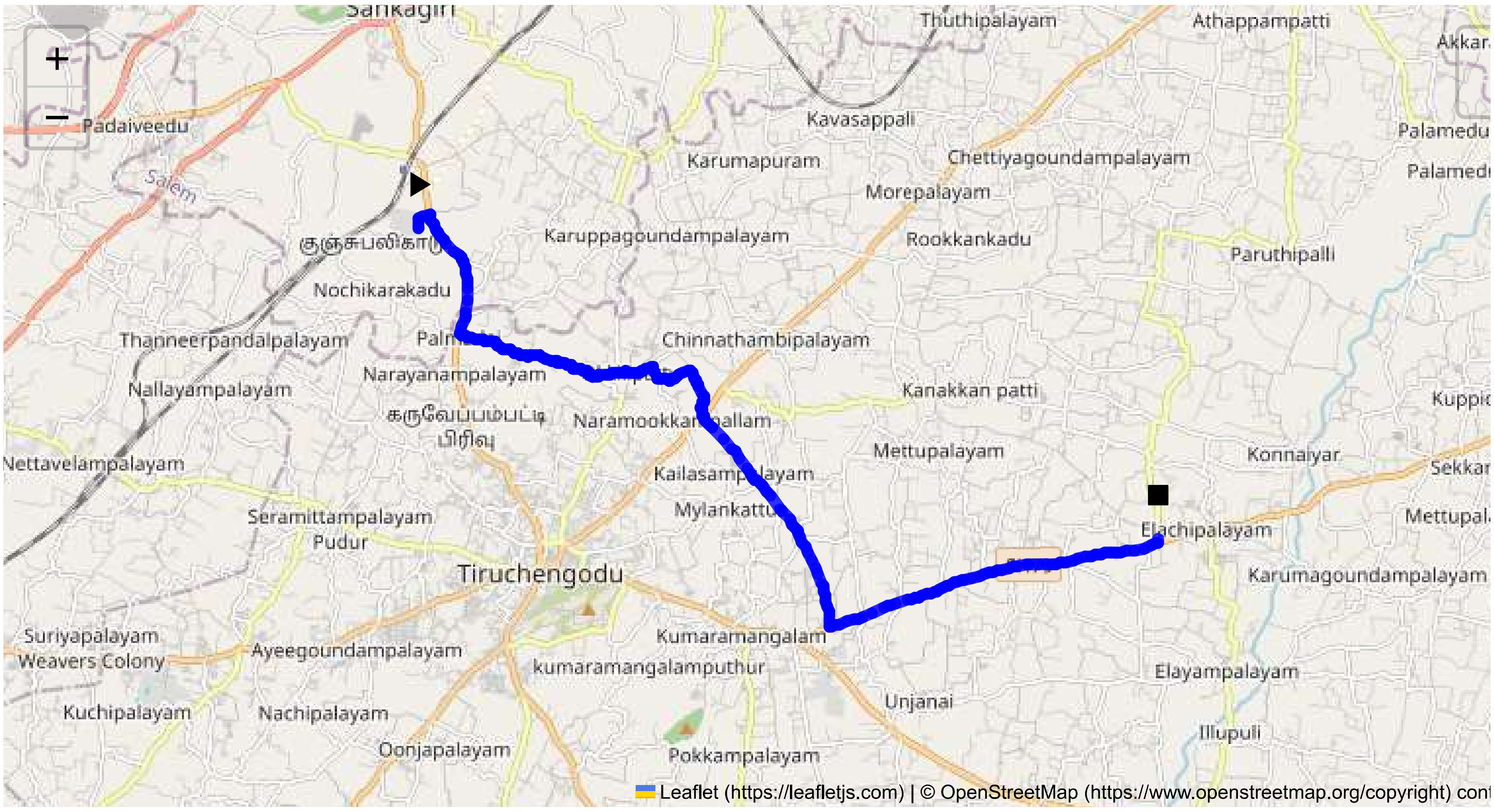
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 20.35 km
Estimated Duration: 0.5 hours
Adjusted Duration (Heavy Vehicle): 0.7 hours
Start: (11.4381, 77.8734)
End: (11.384455, 78.003014)



Welcome to the Journey Risk Management Study

Route Safety Analysis for CVQF+23W, Sangagiri to 92M3+V89, Elachipalayam, Tamil Nadu

1. **Overview of the Route Map:**
   - The route covers approximately 20.35 km in the western region of Tamil Nadu, taking you through semi-urban and rural areas. The primary roads involved are state highways and local roads with varying densities of traffic and infrastructure quality.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:**
   - Tamil Nadu generally experiences a tropical climate. The weather is typically hot and humid with significant rainfall during the monsoon season from October to December. Potential hazards include reduced visibility during heavy rains and the possibility of waterlogged roads, particularly in poorly drained rural areas.
3. **Traffic Patterns:**
   - Traffic can become congested during peak hours, typically from 8:00-10:00 AM and 5:00-8:00 PM near urban centers like Sangagiri. Rural roads might pose issues like slow-moving agricultural vehicles. Watch for congestion spots near major intersections and marketplaces.
4. **Assessment of Road Quality and Infrastructure:**
   - Mixed road conditions can be expected. Urban outskirts may have well-maintained roads, while rural sections might suffer from wear and tear, with potholes and narrow stretches being common. Road signs and lighting can be inconsistent, demanding extra caution, especially during night drives.
5. **Suggestions for Alternative Routes for Emergencies:**
   - Consider local knowledge for quick detours using village roads, but for clarity and assurance, the primary alternative involves slightly backtracking to the nearest state highway and rerouting around congested or hazardous areas.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:**
   - Tamil Nadu follows national guidelines on the transport of hazardous materials, which include restrictions on routes, time-specific constraints, and required documentation. Ensure compliance with all relevant permits and licenses.
7. **Overview of Historical Incidents:**
   - While specific data for this exact route may not be available, common incidents in the region include rollovers and collisions attributed to narrow roads and speeding. Review local news archives for recent episodes to better prepare for known trouble spots.
8. **Environmental Considerations and Sensitive Areas:**
   - Watch for agricultural zones and near villages where environmental sensitivity necessitates cautious driving to prevent contamination and ensure safety. Avoidance of spills and leaks is critical in these sensitive zones.
9. **Analysis of Communication Coverage:**
   - Communication coverage is generally reliable near major towns but can become sporadic in rural regions and certain stretches through forests. Carry mobile devices with multiple network capabilities to ensure connectivity.
10. **Estimated Emergency Response Times:**
    - Response times vary due to proximity to urban centers; nearer to Sangagiri, response might be within 30 minutes, while rural areas can see delays of up to an hour or more due to limited emergency services.
11. **Overall Summary of Risk Assessment:**

- The route presents moderate risk, primarily due to unpredictable weather during monsoons, variance in road quality, and traffic conditions. Ensuring proactive communication, strict adherence to regulations, and caution during inclement weather can help mitigate risks.

Final Recommendations:

- Equip vehicles with all-weather tires and GPS systems for accurate location tracking.
- Maintain constant communication with dispatch for real-time updates.
- Engage in routine checks of safety regulations and emergency protocols.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Blind Spot	Blind Spot	11.41975, 77.88075	10 KM/Hr
5	Turn	Medium	11.41876, 77.88487	30 KM/Hr
6	Turn	Medium	11.41829, 77.88568	30 KM/Hr
7	Turn	Medium	11.41849, 77.88613	30 KM/Hr
8	Turn	Medium	11.41861, 77.88687	30 KM/Hr
9	Turn	Medium	11.41852, 77.88700	30 KM/Hr
10	Turn	Medium	11.41769, 77.88730	30 KM/Hr
11	Turn	Medium	11.41377, 77.90073	30 KM/Hr
12	Turn	Medium	11.41227, 77.90470	30 KM/Hr
13	Turn	Medium	11.41235, 77.90492	30 KM/Hr
14	Turn	Medium	11.41271, 77.90516	30 KM/Hr
15	Turn	Medium	11.41329, 77.90978	30 KM/Hr
16	Turn	High	11.41313, 77.91057	15 KM/Hr
17	Turn	High	11.41307, 77.91058	15 KM/Hr
18	Turn	Medium	11.41334, 77.91266	30 KM/Hr
19	Turn	High	11.41424, 77.91433	15 KM/Hr
20	Turn	Medium	11.41409, 77.91458	30 KM/Hr
21	Turn	Medium	11.41208, 77.91551	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
22	Turn	Medium	11.41205, 77.91556	30 KM/Hr
23	Turn	Medium	11.41207, 77.91604	30 KM/Hr
24	Turn	High	11.41204, 77.91610	15 KM/Hr
25	Turn	Medium	11.41216, 77.91626	30 KM/Hr
26	Turn	Medium	11.41219, 77.91661	30 KM/Hr
27	Turn	Medium	11.41183, 77.91755	30 KM/Hr
28	Turn	Medium	11.41342, 77.92087	30 KM/Hr
29	Blind Spot	Blind Spot	11.41350, 77.92093	10 KM/Hr
30	Turn	Medium	11.41347, 77.92096	30 KM/Hr
31	Turn	Medium	11.41336, 77.92130	30 KM/Hr
32	Turn	High	11.40566, 77.92355	15 KM/Hr
33	Turn	Medium	11.40557, 77.92343	30 KM/Hr
34	Turn	Medium	11.40555, 77.92342	30 KM/Hr
35	Blind Spot	Blind Spot	11.36920, 77.94568	10 KM/Hr
36	Turn	High	11.38375, 78.00328	15 KM/Hr

Route Photos of Risky Spots

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.41975, 77.88075

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41876, 77.88487

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41829, 77.88568

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41849, 77.88613

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41861, 77.88687

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41852, 77.88700

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41769, 77.88730

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41377, 77.90073

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41227, 77.90470

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41235, 77.90492

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41271, 77.90516

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41329, 77.90978

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41313, 77.91057

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41307, 77.91058

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41334, 77.91266

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41424, 77.91433

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41409, 77.91458

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41208, 77.91551

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41205, 77.91556

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41207, 77.91604

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41204, 77.91610

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41216, 77.91626

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41219, 77.91661

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41183, 77.91755

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41342, 77.92087

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.41350, 77.92093

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41347, 77.92096

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41336, 77.92130

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.40566, 77.92355

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.40557, 77.92343

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.40555, 77.92342

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.36920, 77.94568

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38375, 78.00328